History and Ecology of R

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SPE 2026, Tartu

Co-funded by
the European Union

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Pre-history

Before there was R, there was S.

BELL LABORATORIES · MURRAY HILL, N.J. | A COMMEMORATIVE | STATISTICS & DATA ANALYSIS

MAY 5 · 1976 — MAY 5 · 2026

Fifty Years of S

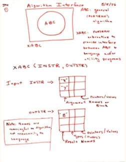
On the half-century of a language that **forever altered** *how we analyze, visualize, and compute with data.*

✦ ✦ ✦

JMC

> *It marks fifty years since Rick Becker and I presented a new project to a few of our colleagues in Statistics and Data Analysis research at Bell Labs. It had no name and we were unsure what it was, exactly. A few months later it was named S, initially called an interactive environment. Only a revised version was referred to as a language. Lately I've been labelling it* `R Version 0`.
>
> — JOHN M. CHAMBERS, RECOLLECTING THE FIRST MEETING · APRIL 2026

EXHIBIT I · THE FIRST VU GRAPH — ALGORITHM INTERFACE · 5/5/76



JMC · 5/5/76

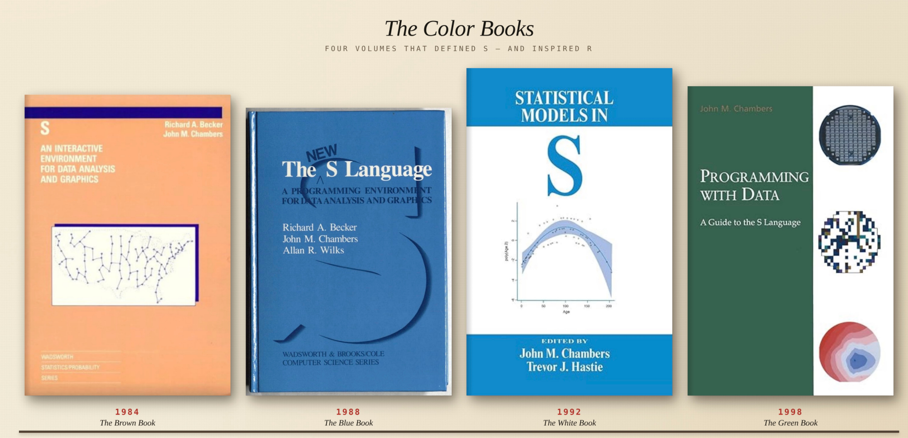
"Some functions that could be called interactively and were implemented as interfaces to Fortran subroutines, along with a general data structure for arguments and other objects." — *Initialed and dated, per Bell System practice.*

ORIGINS — A BRIEF HISTORY

What began in a small conference room at Bell Labs — no name, no formal approval, no certainty of what it was — would grow, over half a century, into the lingua franca of statistical computing. First an interactive environment, then a language, then the foundation of R: an open-source reimplementation that now powers scientific discovery the world over.

EVOLUTION — A TIMELINE

1976	**Conception.** Chambers & Becker sketch the Algorithm Interface on May 5.
1977	**Named S.** *Computational Methods for Data Analysis* is published.
1984	The **Brown Book** introduces S to the world beyond Bell Labs.
1988	The **Blue Book** defines the "New S" — the direct lineage of R.
1993	Ihaka & Gentleman, in Auckland, begin writing **R**.
1999	The ACM awards Chambers its Software System Award — beside UNIX, TeX, the Web.
2026	**Fifty years on.** A language still turning ideas into software — quickly and faithfully.


The Color Books
FOUR VOLUMES THAT DEFINED S — AND INSPIRED R
S
Richard A. Becker
John M. Chambers
AN INTERACTIVE ENVIRONMENT FOR DATA ANALYSIS AND GRAPHICS
WADSWORTH STATISTICS/PROBABILITY SERIES
The NEW S Language
A PROGRAMMING ENVIRONMENT FOR DATA ANALYSIS AND GRAPHICS
Richard A. Becker
John M. Chambers
Allan R. Wilks
WADSWORTH & BROOKS/COLE COMPUTER SCIENCE SERIES
STATISTICAL MODELS IN S
EDITED BY
John M. Chambers
Trevor J. Hastie
John M. Chambers
PROGRAMMING WITH DATA
A Guide to the S Language
1984
The Brown Book
1988
The Blue Book
1992
The White Book
1998
The Green Book

Functions and objects

Everything in S is an **object**.

- The data you input to a statistical model is an object.
- The *output* from a model is also object.

Applying **functions** to objects to create more objects.

- You can chain together a series of function calls to make a powerful analysis pipeline.

Data frames and model formulae

A **data frame** collects observations together in a single object

- Variables are rows
- Individual observations are columns

A **model formula** describes the statistical relationships between variables in a compact way

Programming with Data

> *"We wanted users to be able to begin in an interactive environment, where they did not consciously think of themselves as programming. Then as their needs became clearer and their sophistication increased, they should be able to slide gradually into programming." – John Chambers, Stages in the Evolution of S*

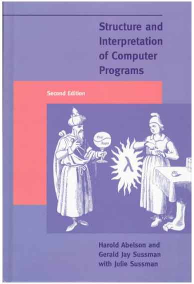
This philosophy was later articulated explicitly in *Programming With Data* (Chambers, 1998) as a kind of mission statement for S

> *To turn ideas into software, quickly and faithfully*

History

How R started, and how it turned into an S clone

The Dawn of R



- Ross Ihaka and Robert Gentlemen at the University of Auckland
- An experimental statistical environment
- Scheme interpreter with S-like syntax
  - Replaced scalar type with vector-based types of S
  - Added lazy evaluation of function arguments
- Announced to *s-news* mailing list in August 1993.

A free software project

- June 1995. Martin Maechler (ETH, Zurich) persuades Ross and Robert to release R under GNU Public License (GPL)
- March 1996. Mailing list *r-testers* mailing list
  - Later split into three *r-announce*, *r-help*, and *r-devel*.
- Mid 1997. Creation of *core team* with access to central repository (CVS)
  - Doug Bates, Peter Dalgaard, Robert Gentleman, Kurt Hornik, Ross Ihaka, Friedrich Leisch, Thomas Lumley, Martin Maechler, Paul Murrell, Heiner Schwarte, Luke Tierney

The draw of S

"Early on, the decision was made to use S-like syntax. Once that decision was made, the move toward being more and more like S has been irresistible"
– Ross Ihaka, R: Past and Future History (Interface '98)

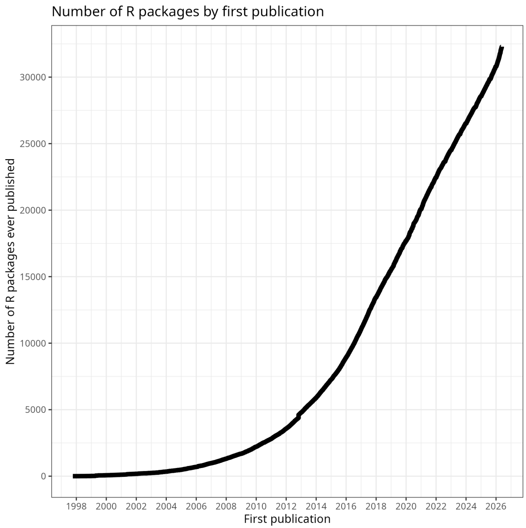
R 1.0.0, a complete and stable implementation of S version 3, was released in 2000.

A Souvenir

Packages

- Comprehensive R Archive Network (CRAN) started in 1997
  - Quality assurance tools built into R
  - Increasingly demanding with each new R release
- Recommended packages distributed with R
  - Third-party packages included with R distribution
  - Provide more complete functionality for the R environment
  - Starting with release 1.3.0 (completely integrated in 1.6.0)

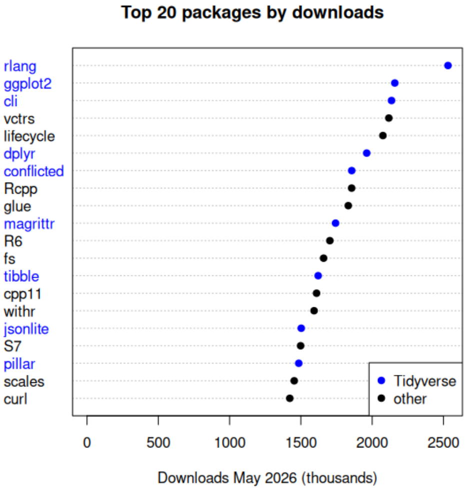
Growth of CRAN



Community

- useR! Annual conference
  - Salzburg (2024), Duke University NC (2025), Warsaw (2026)
- R Journal (`http://journal.r-project.org`)
  - Journal of record, peer-reviewed articles, indexed
  - Journal of Statistical Software (JSS) has many articles dedicated to R packages (`http://jstatsoft.org`)
- Migration to social media
  - Stack Exchange/Overflow, Github, Twitter/X, Mastodon (#rstats)
  - Follow @r-Foundation.bsky.app on BlueSky, or @R_Foundation@fosstodon.org on Mastodon

Much important R infrastructure is now in package space



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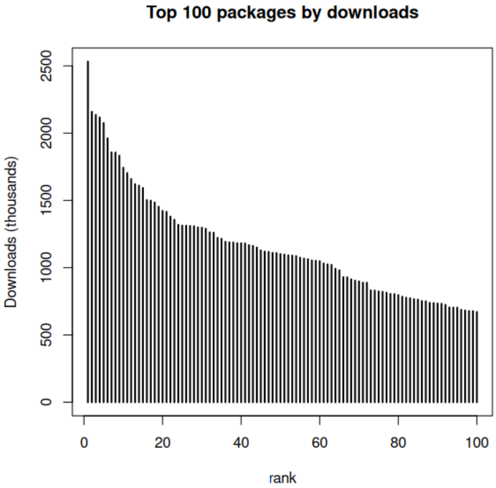

The tidyverse

- Many of the popular packages on CRAN come from the company Posit Software, PBC (formerly R Studio).
- These packages are known as the "tidyverse" (`https://www.tidyverse.org`).
- All packages in the tidyverse have a common design philosophy and work together. Common features are:
  - Non-standard evaluation rules for function calls.
  - Use of the pipe operator |> (or %>%) to pass data transparently from one function call to another.
- The CRAN meta-package `tidyverse` installs all of these packages.

The R Foundation for Statistical Computing

A non-profit organization working in the public interest, founded in 2002 in order to:

- Provide support for the R project and other innovations in statistical computing.
- Provide a reference point for individuals, institutions or commercial enterprises that want to support or interact with the R development community.

The R Consortium

In 2015, a group of organizations created a consortium to support the R ecosystem.

Current members (May 2026)

R Foundation A statutory member of The R Consortium

Gold members Microsoft, Posit

Silver members ASA, esri, GSK, Johnson & Johnson, Merck, Novartis, Novo Nordisk, Pfizer, Swiss Re Procogia, Sanofi, Swiss Re

Local R User Groups

The R Consortium sponsors local groups.

See https://www.meetup.com/pro/r-user-groups/.

R Forwards

- R Forwards is an R Foundation task for to widen the participation of women and other under-represented groups.
- The task force was set up by the R Foundation in December 2015 to address the underrepresentation of women and rebranded in January 2017 to accommodate more under-represented groups such as LGBT, minority ethnic groups, and people with disabilities in the R community.
- See `https://forwards.github.io`

R Ladies

- As a diversity initiative, the mission of R-Ladies is to achieve proportionate representation by encouraging, inspiring, and empowering people of genders currently under-represented in the R community. R-Ladies' primary focus, therefore, is on supporting minority gender R enthusiasts to achieve their programming potential, by building a collaborative global network of R leaders, mentors, learners, and developers to facilitate individual and collective progress worldwide.
- See `https://rladies.org`

R Contribution Working Group

- Initiative to encourage new contributors to R core, with a focus on diversity and inclusion.
- Includes a developer guide aimed at getting new developers on board.
- See `https://contributor.r-project.org`

What does all of this mean for the course?

- R incorporates 50 years of ideas in statistical computing from multiple contributors.
- There is usually more than one way to do something in R.
- Some of the peculiarities of the R language are there for historical reasons.
- The course does not cover some of the recent additions to the R ecosystem.

Resources

- Chambers J, Stages in the Evolution of S
- Becker, R, A Brief History of S
- Chambers R, Evolution of the S language
- Ihaka, R and Gentleman R, R: A language for Data Analysis and Graphics, *J Comp Graph Stat*, **5**, 299–314, 1996.
- Ihaka, R, R: Past and Future History, Interface 98.
- Ihaka, R, Temple Lang, D, Back to the Future: Lisp as a Base for a Statistical Computing System
- Fox, J, Aspects of the Social Organization and Trajectory of the R Project, R Journal, Vol 1/2, 5–13, 2009.